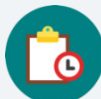


12.8 General Form of a Circle

Lesson Introduction

 Benchmarks	MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features. MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.			 Learning Targets	- I can rewrite the equation of a circle from general form to center-radius form. - I can determine key features of a circle given the equation in general form. - I can write the domain and range using inequality notation, interval notation, or set-builder notation.
 Benchmark Coherence	Building On: MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable, quadratic equations over the real number system. Within this benchmark, the expectation is to solve by factoring techniques, taking square roots, the quadratic formula, and completing the square.			Working Toward: N/A	
 Essential Questions	- How can I convert to general form of a circle when it is given in center-radius form? - How do I use completing the square to rewrite the equation of a circle?				
 Lesson Narrative	Students begin to see the equation of a circle in general form. This lesson activates prior knowledge regarding completing the square to show how to convert from general form of a circle to center-radius form of a circle. Students will be able to convert from either form of a circle and identify the key features of a given circle.				
 Component Summary	12.8.1 Warm-Up (~5 min.)	Conic sections discovery (SE p. 267)	12.8.4 Guided Instruction (10 min.)	Conversion from the general form to center-radius form (SE p. 270)	
	12.8.2 Collaboration (10 min.)	Discovering general form (SE p. 267)	12.8.5 Guided Practice (10 min.)	Converting forms of a circle (SE p. 271)	
	12.8.3 Refresher (10-15 min.)	Completing the square (SE p. 268)	12.8.6 Wrap-Up (~5 min.)	Self-Reflection on understanding of key concepts (SE p. 272)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - General Form <u>Additional Terms:</u> - Center-Radius Form		(Optional) Algebra Tiles		12.8.3 Refresher reviews completing the square. Consider preparing your classroom-specific vocabulary or glossary strategies for support (Word Wall, Math Cards, etc.).

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may have forgotten algebraic concepts such as completing the square or factoring perfect square trinomials to convert the circle equation from center-radius form to general form. - Students may misunderstand key or additional terms. - Students may miscalculate when completing the square or factoring.	- Review perfect squares and how to simplify square roots. - Give students several examples of how to graph a circle, given its equation in general form, and how to find the equation of a circle, given its graph. Be sure to give examples of circles whose center coordinates are both negatives, and whose center coordinates are a combination of both positive and negative, as well as examples of circles whose center coordinates are at the origin or are on one of the axes. - Determine if your students need the support of all or part of the 12.8.3 Refresher.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time Consider using this instructional routine throughout this lesson. Give students a few minutes to complete questions individually. Then, allow students time to talk with a partner about their circle and equation. This will allow students to ask clarifying questions, if needed, and have a person to share their ideas with. Students will then fix their equation of a circle, if needed. Consider having students do this throughout the lesson so students can get stronger with writing equations of a circle in general form.	MA.K12.MTR.5.1: Patterns and Structure 12.8.3 Refresher reviews completing the square, which is essential in converting equations of circles in center-radius form to general form. Mathematicians who use patterns and structure to help understand and connect mathematical concepts focus on relevant details within a problem. While competing 12.8.3 Refresher, encourage students to see the patterns and structure to help them understand and connect the completing the square concept. This supports students in developing generalizations based on the similarities found among problems.
 Differentiate Instruction	Consider using additional visuals, such as algebra tiles, to help students better conceptualize the concept completing the square to rewrite the equation of a circle for visual and kinesthetic entry-point. Students will have the chance to draw their own conclusions and discuss their rationale with peers, providing an opportunity for auditory learners to verbalize their understanding. Students can also utilize Study Expert videos from Algebra 1 lessons covering MA.912.AR.3.1 and completing the square for additional scaffolding (in Algebra 1 Unit 8 Lessons 5-7).	
Lesson Notes:		

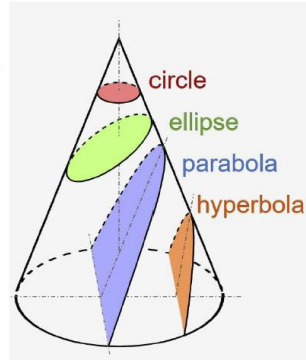
12.8.1 Warm-Up: Conic Sections

Component Overview: Students begin constructing meaningful entry-points into mathematical discussions on the concept of conic sections.



12.8.1 Warm-Up: Conic Sections

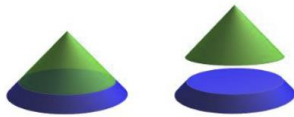
Conic sections are the family of curves created when a plane slices through a cone. A circle is best known as a conic section. A diagram of five conic sections is shown.



According to Greek philosopher Proclus, Menaechmus, a member of Plato's Academy, discovered conic sections around 350 B.C. Approximately 100 years later, Apollonius, a Greek geometer and astronomer, developed and published the theory of conic sections. His work earned him the title of the Great Geometer.



Consider displaying cone, circle, ellipse, parabola, and hyperbola with interactive digital manipulatives, such as GeoGebra.



1. Use the diagram to explain why when a cone is sliced horizontally the result is a circle.
Answers will vary. A horizontal slice is parallel to the base and will have the same shape as the base, which is a circle.

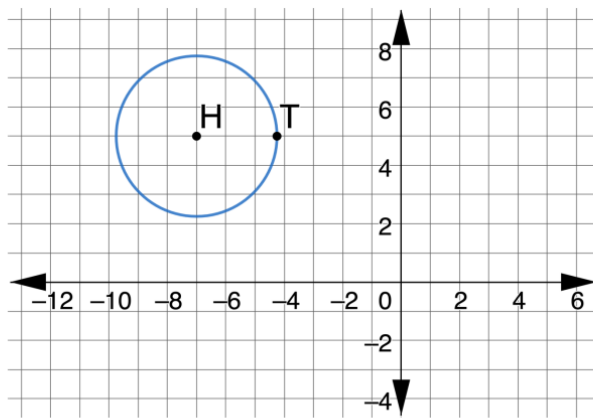
12.8.2 Collaboration: General Form

Component Overview: Students are writing the equation of a circle in center-radius form using the given center and point. Students are then rewriting the equation of the circle in general form.



12.8.2 Collaboration: General Form

- The center of circle H is located at $(-7, 5)$. Point T lies on circle H at the point $(-4.25, 5)$.



- Write the equation of the circle in center-radius form, $(x - h)^2 + (y - k)^2 = r^2$.
 $(x + 7)^2 + (y - 5)^2 = 7.5625$ or
 $(x + 7)^2 + (y - 5)^2 = 2.75^2$
- Expand each term of the center-radius form to write an equivalent expression.

Term	Equivalent Expression
$(x + 7)^2$	$x^2 + 14x + 49$
$(y - 5)^2$	$y^2 - 10y + 25$
$(2.75)^2$	7.5625



The **general form** of an equation of a circle is $ax^2 + by^2 + cx + dy + e = 0$.

- Use the equivalent expressions of the center-radius form to rewrite the center-radius form of circle H in general form.
 $x^2 + y^2 + 14x - 10y + 66.4375 = 0$



As students collaborate, listen for opportunities to prompt mathematical discussion. Elicit any exemplar student responses whole-group after the component.

- If a coordinate in the center is positive, will the equation always, sometimes, or never have a subtraction sign in front of the positive value? Explain.
- How can you find the radius of the circle?
- Can you think of another way to find the radius of the circle? (Distance Formula or plugging in point T into the equation.)



Ask your students to explain what information is needed in order to write the equation of a circle. Have them explain how you could verify your equation of a circle is correct.

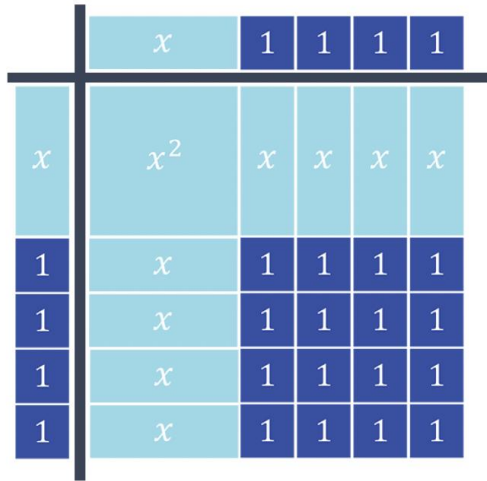
12.8.3 Refresher: Completing the Square

Component Overview: Students review completing the square using a pictorial representation of algebra tiles. Students are asked to complete the square by finding the perfect square trinomial. This component will allow students to be able to go from general form to center-radius form.



12.8.3 Refresher: Completing the Square

The binomial $(x + 4)^2$ can be rewritten as $x^2 + 8x + 16$, which is a perfect square trinomial. The algebra tiles represent both $(x + 4)^2$ and $x^2 + 8x + 16$.



1. Discuss with your partner how the diagram shows that $x^2 + 8x + 16$ is a perfect square trinomial. Summarize your discussion.

Answers may vary.

The shape of the algebra tiles make a square.

2. Discuss with your partner how the diagram shows that $x^2 + 8x + 16 = (x + 4)^2$.

The outsides of the algebra tiles show that $(x + 4)(x + 4) = (x + 4)^2$.



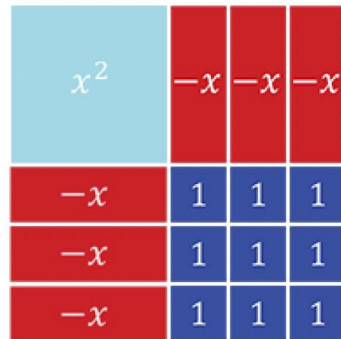
Students must be able to complete the square to effectively engage in 12.8.4 Guided Instruction. 12.8.3 Refresher provides embedded scaffolding for students to review their working knowledge of MA.912.AR.3.1 and completing the square.



Consider splitting the class into groups for this component and pulling a small group for students that will struggle with this content.

12.8.3 Refresher: Completing the Square (continued)

3. Write the perfect square trinomial and the binomial squared that are represented by the diagram.

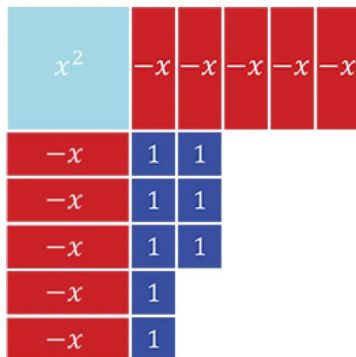


$$x^2 - 6x + 9 = (x - 3)^2$$



Provide hands-on or digital manipulatives for students to engage with algebra tiles for kinesthetic and visual entry-points.

4. The trinomial $x^2 - 10x + 8$ is shown.



- a. How many 1 tiles have to be added to complete the square?

17

12.8.3 Refresher: Completing the Square (continued)

Completing the square by adding 1 tiles will change the value of the original trinomial unless the same number of -1 tiles are also added.

b. Complete the table.

Completing the Square	
Original Expression	$x^2 - 10x + 8$
Completing the square	$x^2 - 10x + 8 + 17$ number of 1 tiles added
Adding the opposite so the expression is equivalent to the original expression	$x^2 - 10x + 8 + 17 + -17$ number of -1 tiles added
Rewriting so that there is a perfect square trinomial	$(x^2 - 10x + 25) - 17$ perfect square trinomial
Writing the perfect square trinomial as a binomial squared	$(x - 5)^2 - 17$



Make a meaningful connection between 12.8.3 Refresher and 12.8.4 Guided Instruction. Why are we reviewing something that we learned in Algebra 1? The learning experience of this lesson should remain centered on the concept of converting between center-radius form and general form.

Students who are able to articulate why completing the square matters in the equations of circles may struggle with the algebraic operation of completing the square, but should be recognized for their learning from the lesson.

12.8.4 Guided Instruction: Rewriting the General Form of the Equation of a Circle

Component Overview: Students are completing the square to rewrite the equation of a circle from general form to center-radius form. Students are identifying the center and radius of each equation of a circle.



12.8.4 Guided Instruction: Rewriting the General Form of the Equation of a Circle

There are several ways to mathematically represent the domain and range.

Written Description	Inequalities	Set-Builder Notation	Interval Notation
All real numbers greater than or equal to -5 and less than or equal to 5	$-5 \leq x \leq 5$	$\{x: -5 \leq x \leq 5\},$ $\{x -5 \leq x \leq 5\},$ $\{x: x \geq -5 \text{ and } x \leq 5\},$ or $\{x x \geq -5 \text{ and } x \leq 5\}$	$[-5, 5]$

Rewrite each general form equation of a circle to center-radius form of a circle. Then, give the key features of the circle. Write the domain and range using interval notation and set-builder notation.

1. $x^2 + y^2 - 14x + 6y + 26 = 0$
 $(x^2 - 14x + 49) + (y^2 + 6y + 9) = -26 + 49 + 9$
 $(x - 7)^2 + (y + 3)^2 = 32$

Center-Radius Form	Center	Radius
$(x - 7)^2 + (y + 3)^2 = 32$	$(7, -3)$	$\sqrt{32}$ or $4\sqrt{2}$
	Domain	Range
	$\{x: (7 - \sqrt{32}) \leq x \leq (7 + \sqrt{32})\}$ $[(7 - \sqrt{32}), (7 + \sqrt{32})]$	$\{y: (-3 - \sqrt{32}) \leq y \leq (-3 + \sqrt{32})\}$ $[(-3 - \sqrt{32}), (-3 + \sqrt{32})]$



Provide algebra tiles for additional entry-points when completing the square.



Make connections between 12.8.3 Refresher and 12.8.4 Guided Instruction so these activities do not feel isolated:

- How does this equation look the same as what we just did in 12.8.3 Refresher?
- How does it look different?
- Why might we need to complete the square here?



You may need to refresh students on how to write square roots in simplest form for the radius.

12.8.4 Guided Instruction: Rewriting the General Form of the Equation of a Circle (continued)

2. $x^2 + y^2 - 18x - 10y = -52$
 $(x^2 - 18x + 81) + (y^2 - 10y + 25) = -52 + 81 + 25$
 $(x - 9)^2 + (y - 5)^2 = 54$

Center-Radius Form	Center	Diameter
$(x - 9)^2 + (y - 5)^2 = 54$	$(9, 5)$	$2(\sqrt{54})$ or $6\sqrt{6}$
	Domain	Range
	$\{x: (9 - \sqrt{54}) \leq x \leq (9 + \sqrt{54})\}$ $[(9 - \sqrt{54}), (9 + \sqrt{54})]$	$\{y: (5 - \sqrt{54}) \leq y \leq (5 + \sqrt{54})\}$ $[(5 - \sqrt{54}), (5 + \sqrt{54})]$

12.8.5 Guided Practice: Converting Forms of a Circle

Component Overview: Students begin to formalize the process of converting forms of a circle. This component gives students the opportunity to apply the conversion to both center-radius form and general form.



12.8.5 Guided Practice: Converting Forms of a Circle

1. Consider the equation of the circle represented by $(x + 8)^2 + (y + 11)^2 = 14^2$.
 - a. Rewrite $(x + 8)^2 + (y + 11)^2 = 14^2$ in general form.
 $x^2 + y^2 + 16x + 22y - 11 = 0$
 - b. Give the key features of $(x + 8)^2 + (y + 11)^2 = 14^2$

Center	Diameter	Radius
$(-8, -11)$	28	14
Domain	Range	
$-22 \leq x \leq 6$	$-25 \leq y \leq 3$	

2. Consider the equation of the circle represented by $x^2 + y^2 - 20x = -36$.
 - a. Rewrite $x^2 + y^2 - 20x = -36$ in center-radius form.
 $(x^2 - 20x + 100) + y^2 = -36 + 100$
 $(x - 10)^2 + y^2 = 64$
 - b. Give the key features of $x^2 + y^2 - 20x = -36$

Center	Diameter	Radius
$(10, 0)$	16	8
Domain	Range	
$[2, 18]$	$[-8, 8]$	



Think-Pair-Share

Consider conducting the Think-Pair-Share instructional routine for 12.8.5 Guided Practice. Students will make better sense of thinking and reasoning with the opportunity to independently process then refine answers with a partner, before the teacher elicits more precise responses as a whole group.



Use inquiry as students work to synthesize and apply their learnings from 12.8.3 Refresher and 12.8.4 Guided Instruction:

- What form of the circle is given? How do you know?
- How can you convert from center-radius form to general form?
- How can you identify the center, diameter, and radius of a circle?



Clarify, Critique, Correct

Consider the Clarify, Critique, and Correct instructional routine as a way to deepen your students' understanding. Have the students exchange their work for these two problems to analyze, reflect, and develop their partner's work. This will prompt student reflection, whether it is for students to improve upon, correct errors, or clarify their partner's work.

12.8.6 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of the content of this lesson.



12.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Rewriting center-radius form to general form			
Key features of a circle			
Completing the square			
Rewriting general form to center-radius form			
Set-builder notation			
Interval notation			

Answers may vary.



Note that completing the square is only one of four targets within 12.8.6 Wrap-Up. During the lesson, continue centering students' learning experiences on the concept of changing from center-radius to general form, rather than getting stuck on the algebraic procedure of completing the square.